

3 Partial Derivatives

3.1 Functions of Several Variables

Definition 3.1.1 A function of two variables is a rule which assigns a number to each ordered pair of real numbers (x, y) from a given domain D , which is denoted $f(x, y)$. The set D is known as the domain, and its range is the set of real values obtained as output of f .

Example 3.1.2 For the following function, evaluate $f(3, 2)$ and find and sketch the domain.

$$f(x, y) = \frac{\sqrt{x + y + 1}}{x - 1}.$$

Example 3.1.3 Find the domain of the function

$$f(x, y) = \sqrt{9 - x^2 - y^2}.$$

3.1.1 Graphs

Definition 3.1.4 If f is a function of two variables with domain D , then the graph of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and (x, y) is in D .

Example 3.1.5 Sketch the graph of the function $f(x, y) = 6 - 3x - 2y$.

Example 3.1.6 Draw several level curves of the function and sketch the graph of the function.

$$f(x, y) = 4x^2 + y^2 + 1.$$

Example 3.1.7 Draw several level curves of the given function and sketch the graph of the function.

$$f(x, y) = \sqrt{4 - 4x^2 - y^2}.$$